Issuing(Revision) Date : 09-05-2023
Version number : 1

Safety Data Sheet(SDS)

According to Regulation (EC) No. 2020/878

SECTION 1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product identifier	:	PLASTICIZER GL300 BULK
Other means of identification	:	
PCN No.	:	

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses	:	48.Others (Sheet, Gloves, Wallpaper, Toy)
Uses advised against	:	

1.3 Details of the supplier of the safety data sheet

Name	:	LG Chem, Ltd.
Address	:	54, Naju-ro, Naju-si, Jeollanam-do, Republic of Korea
Telephone number	:	
Fax number	:	
Email	:	

1.4 Emergency telephone number

Emergency telephone number	:	+49-69-710-455-138 Customer Solution Center Europe
Opening hours	:	09:00~17:00 (CET, Central European Time)
Other comments(e.g. language(s) of the phone	:	English, Deutsch, Korean available.

SECTION 2. Hazards identification

2.1 Classification of the substance or mixture according to Regulation (EC) No 1272/2008

- Not classified

2.2 Label elements

Hazard pictogram

The product does not require a hazard warning label in accordance with GHS criteria.

Labelling according to Regulation (EC) No 1272/2008 [CLP]

Signal word

- NONE

Hazard statements

- Not required

Precautionary statements

-Not required

2.3 Other hazards

- According to Annex XIII of (EC) No 1907/2006, the substance does not meet PBT or vPvB criteria.
- According to Regulation(EU) 2017/2100 and 2018/605, the substance does not affect to endocrine system.
- The substance is not listed in Article 59
- No other hazards have been identified

SECTION 3. Composition/information on ingredients

3.1. Substances

Not applicable

3.2 . Mixtures

Substance name	CAS No. EC No. EU REACH No.	Classification	SCL M-Factor	ATE	PCT(wt%)

2-Ethylhexanol	104-76-7 203-234-3 	No data available	No data available No data available	Acute toxicity(Oral) : 2774mg/kg, Acute toxicity(Dermal) : 1970mg/kg	0.03
1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester	6422-86-2 229-176-9 	No data available	No data available No data available	No data available	99.97

※ Classification : Classification according Regulation(EC) No. 1272[CLP]
SCL : Specific concentration limit
M-factor : The multiplication factor
ATE : The acute toxicity estimate

SECTION 4. First aid measures

4.1 Description of first aid measures

○4.1.1 Following eye contact
- Get medical aid immediately.

- In case of contact with material, immediately flush eyes with running water for at least 15 minutes.

○4.1.2 Following skin contact
- Get medical aid immediately.

- In case of contact with material, immediately flush skin with running water for at least 15 minutes.

- Launder contaminated clothing and shoes before re-use.

- Remove and isolate contaminated clothing and shoes.

○4.1.3 Following inhalation
- Administer oxygen if breathing is difficult.

- Give artificial respiration if victim is not breathing.

- Move to fresh air.

- Seek immediate medical assistance.

○4.1.4 Following ingestion
- Get medical aid immediately.

- If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

- If unconscious but breathing, never give anything by mouth.

4.2 Most important symptoms and effects, both acute and delayed
- No data available

4.3 Indication of any immediate medical attention and special treatment needed
- Ensure that medical personnel are aware of the material(s) involved and take precautions to

protect themselves.

- No specific antidote. Treatment of exposure should be directed at the control of symptoms and the clinical condition of the patient.

SECTION 5. Firefighting measures

5.1 Extinguishing media

○Suitable extinguishing media
- Large fire: Water spray/fog, regular foam (Suitable extinguishing media).

- Small fire: Dry sand, dry chemical, alcohol-resistant foam, water spray, regular foam, CO2 (Suitable extinguishing media).

○Unsuitable extinguishing media
- High-pressure water (Unsuitable extinguishing media).

5.2 Special hazards arising from the substance or mixture(Hazardous combustion products)

- No data available

5.3 Advice for firefighters

- Dike fire-control water for later disposal; do not scatter the material.

- Fire involving Tanks: ALWAYS stay away from tanks engulfed in fire.

- Fire involving Tanks: Cool containers with flooding quantities of water until well after fire is out.

- Fire involving Tanks: Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.

- Move containers from fire area if you can do it without risk.

- Runoff may cause pollution.

- Substance may be transported hot.

SECTION 6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

6.1.1 For non-emergency personnel
○Emergency procedures
- Removal of ignition sources, provision of sufficient ventilation.

○Protective equipment
- Wash thoroughly after handling.

6.1.2 For emergency responders
- Please note that materials and conditions to be avoided.

6.2 Environmental precautions

- Prevent entry into waterways, sewers, basements or confined areas.

6.3 Methods and material for containment and cleaning up

6.3.1 For containment
- Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers.

6.3.2 For cleaning up
- Clear spills immediately.

- Don't use a brush or compressed air for cleaning surfaces or clothing.

6.3.3 Other information
- No data available

6.4 Reference to other sections
- Section 8 (protective equipment), section 13 (disposal instructions)

SECTION 7. Handling and storage

7.1 Precautions for safe handling
- Avoid any skin and eye contact when insert undiluted solution. Wash ... thoroughly after handling.

- CAUTION: High temperature.

- Caution: Dangerous fire hazard when exposed to heat, or flame, sparks.

- Please note that materials and conditions to be avoided.

- Use adequate machine for prevention when package handling.

- Wash thoroughly after handling.

- Wear an appropriate Personal protection. (See Exposure Controls/Personal Protection section.)

7.2 Conditions for safe storage, including any incompatibilities
- Choose a place that can be protected from strong oxidizers and acid.

- Drum Handling: Must work at safe place., Loading more than 3 stack is prohibited.

- Store containers: AVOID the place where can be damage and contamination.

- Store in a cool/low-temperature, well-ventilated {dry} place {away from heat and ignition sources}

7.3 Specific end uses
- See section 1 for recommended use.

SECTION 8. Exposure controls/personal protection

8.1 Control parameters

Components	Occupational exposure	ACGIH regulations	Biological limit values	DNEL/DMEL	PNEC-Values
1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester	TWA : No data available	TWA : No data available	No data available	No data available	No data available
	STEL : No data available	STEL : No data available			
2-Ethylhexanol	TWA : 1ppm	TWA : A3ppm	No data available	No data available	No data available
	STEL : No data available	STEL : No data available			

8.2 Exposure controls
8.2.1 Appropriate engineering controls
- No data available

8.2.2 Individual protection measures, such as personal protective equipment
○ Eye/face protection

- Wear suitable protective goggles and face shields.

○ Respiratory protection
- Atmospheric levels should be maintained below the exposure guideline. For most conditions, no respiratory protection should be needed; however, if material is heated or sprayed, use an approved air-purifying respirator. Use the following CE approved air-purifying respirator: Organic vapor cartridge with a particulate pre-filter, type AP2.

○ Skin protection
(i) Hand protection
- Use gloves chemically resistant to this material when prolonged or frequently repeated contact could occur. Use chemical resistant gloves classified under Standard EN374: Protective gloves against chemicals and micro-organisms. Examples of preferred glove barrier materials include: Butyl rubber. Polyethylene. Neoprene. Natural rubber ("latex"). Polyvinyl chloride ("PVC" or "vinyl"). Nitrile/butadiene rubber ("nitrile" or "NBR"). Polyvinyl alcohol ("PVA"). Ethyl vinyl alcohol laminate ("EVAL"). When prolonged or frequently repeated contact may occur, a glove with a protection class of 4 or higher (breakthrough time greater than 120 minutes according to EN 374) is recommended. When only brief contact is expected, a glove with a protection class of 1 or higher (breakthrough time greater than 10 minutes according to EN 374) is recommended. NOTICE: The selection of a specific glove for a particular application and duration of use in a workplace should also take into account all relevant workplace factors such as, but not limited to: Other chemicals which may be handled, physical requirements (cut/puncture protection, dexterity, thermal protection), potential body reactions to glove materials, as well as the instructions/specifications provided by the glove supplier.

- Wear suitable protective gloves.

(ii) Other
- No data available

○ Thermal hazards
- Wear appropriate protective clothing considering the physical and chemical properties of chemicals.

8.2.3 Environmental exposure controls
- Ensure not to cause environmental pollution by discharging into rivers or other waterways.

- See section 6

SECTION 9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	Liquid	
Relative Vapour density	13.5 ((air=1))	
Density/Relative density	0.9835 (@ 20/20 ℃)	
Kinematic viscosity	No data available	
Decomposition temperature	No data available	
Auto ignition temperature	No data available	
Partition coefficient(n-octanol/water)	8.39	
Solubility	4 ㎎/ℓ (20℃)	
Vapour pressure	0.0000214 ㎜Hg (25℃)	
Upper/lower flammability or explosive limits	No data available	
Flammability(solid, gas)	No data available	
Flash point	238℃	
Initial boiling point and boiling range	383℃	
Melting point/freezing point	-48℃	

pH	No data available	
Odour	Very weak smell	
Colour	Colorless ~ slightly yellow	
Particle characteristics	No data available	

9.2 Other information

9.2.1 Information with regard to physical hazard classes

Flammable liquids	Initial boiling point and boiling range:383℃, Flash point:238℃	

9.2.2 Other safety characteristics

No data available

SECTION 10. Stability and reactivity

10.1 Reactivity

- Containers may explode when heated.
- Fire may produce irritating and/or toxic gases.
- May cause toxic effects if inhaled.
- Some may burn, but not ignite easily.
- Stable under normal temperatures and pressures.

10.2 Chemical stability

- No hazardous reaction when handled and stored according to provisions.
- Stable under normal temperatures and pressures.

10.3 Possibility of hazardous reactions

- Stable under normal temperatures and pressures.

10.4 Conditions to avoid

- Exposure to elevated temperatures can cause product to decompose. Avoid moisture.

10.5 Incompatible materials

- Avoid contact with: Strong oxidizers.

10.6 Hazardous decomposition products

- No known hazardous decomposition products

SECTION 11. Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

○Acute toxicity
●Acute toxicity(Oral) > PRODUCT : Not classified

- 2-Ethylhexanol : LD50 1516 ~ 2774 ㎎ / ㎏ experimental species: Rat, Source: IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

●Acute toxicity(Dermal) > PRODUCT : Not classified

- 2-Ethylhexanol : LD50 1970 ㎎ / ㎏ experimental species: Rabbit, Source: NLM,THOMSON

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

●Acute toxicity(Inhalation:Gases) > PRODUCT : Not classified

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

●Acute toxicity(Inhalation:Vapours) > PRODUCT : Not classified

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

●Acute toxicity(Inhalation:Dust/mist) > PRODUCT : Not classified

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Skin corrosion/ irritation > PRODUCT : Not classified

- 2-Ethylhexanol : Rabbit / high stimulation, Source: ICULID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : When intermittently exposed skin in humans causes a mild, Source: TOMES;RTECS

○Serious eye damage/ irritation > PRODUCT : Not classified

- 2-Ethylhexanol : Exhibits a moderate to severe irritation of the cornea, iris and conjunctiva in Test (OECD TG 405) is applied 0.1 mL of the test substance in a rabbit eye irritation index was 28.59 / 110 (DFGMAK-Doc 20 (2003)). Corneal haze of the test substance 0.1 mL In another test with a rabbit, after 24 hours application to conjunctival sac, iritis, observing the redness and edema of conjunctiva eye irritation index (MMAS) is 51.3 / 110, 10 to 14, after recovery (ECETOC TR48 (1998)), Source: DFGMAK-Doc 20 (2003), ECETOC TR48 (1998)

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : Since the data on the eye irritation skin irritation but is being considered

○Respiratory or skin sensitisation > PRODUCT : Not classified

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Skin sensitization > PRODUCT : Not classified

- 2-Ethylhexanol : Human unresponsiveness, Source: IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Carcinogenicity > PRODUCT : Not classified

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Germ cell mutagenicity > PRODUCT : Not classified

- 2-Ethylhexanol : In vitro / audio, Source: IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Reproductive toxicity > PRODUCT : Not classified

- 2-Ethylhexanol : By pregnancy 12 days oral administration of a second rat, but is reported to the toxicity of the dams, can see the reception, it caused an increase in deformities, such as the children over the tail limb deformities (DFGMAK-Doc.20 (2003)), in addition, the engine generator of the rat in the oral administration of a developmental toxicity test death in dams, general symptoms, a dose that was available on the feeding amount of depression and weight gain suppression, absorption times, a clear increase in the post-implantation loss, renal expansion or mulnyo in addition to the increase in fetal gwanjeung indicate an increase in skeletal deformities, the material is classified as there are said to have a capacity only teratogenicity causing toxicity in maternal and fetal times, conclusions from (DFGMAK-Doc.20 (2003)), Category 2., Source: DFGMAK-Doc.20 (2003)

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Specific target organ toxicity (single exposure) > PRODUCT : Not classified

- 2-Ethylhexanol : To severely affect the show epic, headache, dizziness, fatigue, although support for the occupation exposure of people are reported to cause mild blood pressure and (PATTY (5th, 2001)), in animal studies the mouse, rat, a guinea pig 1 once inhaled dose tested (1.8 mg / L / 4 hours, mist) death is not in, you can view the central nervous system inhibiting the transfer and the eyes, the nose, but the mucous membrane irritation of the throat and respiratory route recognition, administered after one hour recovered (jFCA 786 (1993), DFGMAK-Doc.20 (2003)) (a) classified as a Category 3 year (anesthetic actions, respiratory tract) based on a report that., Source: PATTY (5th, 2001), JFCA 786(1993), DFGMAK-Doc.20(2003)

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Specific target organ toxicity (repeated exposure) > PRODUCT : Not classified

- 2-Ethylhexanol : Rat / toxic effects observed, Source: IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

○Aspiration hazard > PRODUCT : Not classified

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

11.2. Information on other hazards

○11.2.1. Endocrine disrupting properties

- 2-Ethylhexanol : According to Regulation(EU) 2017/2100 and 2018/605, the substance not affects to endocrine system.

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : According to Regulation(EU) 2017/2100 and 2018/605, the substance not affects to endocrine system.

○11.2.2. Other information

- 2-Ethylhexanol : No other hazards have been identified

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No other hazards have been identified

SECTION 12. Ecological information

12.1 Toxicity > PRODUCT : Not classified

●Fish
- 2-Ethylhexanol : LC50 17.1 ㎎ / ℓ 96 hr Leuciscus idus, Source: Directive 84/449/EEC, C1, GLP, IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

●Crustaceans
- 2-Ethylhexanol : EC50 39 ㎎ / ℓ 48 hr Daphnia magna, Source: Directive 84/449/EEC, C. 2, GLP, IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

●Aquatic Algae
- 2-Ethylhexanol : ErC50 11.5 ㎎ / ℓ 72 hr Scenedesmus subspicatus, Source: Directive 87/302/EEC, C. 2, GLP, IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

12.2 Persistence and degradability

●Persistence
- 2-Ethylhexanol : 2.73 log Kow ((estimated))

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : 8.390 log Kow, Source: National Library of Medicine(NLM)(http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?CHEM)

●Degradability
- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

●Biodegradation
- 2-Ethylhexanol : 55 (%) 17 day, Source: Directive 84/449/EEC, C5, GLP, IUCLID

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

12.3 Bioaccumulative potential

- 2-Ethylhexanol : 13, Source: HSDB

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

12.4 Mobility in soil

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : 870,000, Source: National Library of Medicine/Hazardous Substances Data Bank(NLM/HSDB)(http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?HSDB)

12.5 Results of PBT and vPvB assessment

- 2-Ethylhexanol : NOT_APPLICABLE

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : NOT_APPLICABLE

12.6 Endocrine disrupting properties

- 2-Ethylhexanol : According to Regulation(EU) 2017/2100 and 2018/605, the substance not affects to endocrine system.

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : According to Regulation(EU) 2017/2100 and 2018/605, the substance not affects to endocrine system.

12.7 Other adverse effects > PRODUCT : Not classified

- 2-Ethylhexanol : No data available

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester : No data available

SECTION 13. Disposal considerations

13.1 Waste treatment methods

13.1.1 Product / Packaging disposal
- Empty containers should be taken to an approved waste handling site for recycling or disposal.

Waste codes / waste designations according to LoW
- 13-08 oil wastes not otherwise specified

13.1.2 Waste treatment-relevant information
- Disposal according to local regulations.

13.1.3 Sewage disposal-relevant information
- Disposal according to local regulations and avoid release to the environment.

13.1.4 Other disposal recommendations
- No data available.

SECTION 14. Transport information

14.1 UN number or ID number : Not applicable

14.2 UN proper shipping name : Not applicable

14.3 Transport hazard class(es) : Not applicable

14.4 Packing group : Not applicable

14.5 Environmental hazards : Not applicable

14.6 Special precaution for user

Emergency measures in case of fire : Not applicable

Emergency measures in the effluent : Not applicable

14.7 Maritime transport in bulk according to IMO instruments

- **ADR**
  - •**Tunnel restriction code** : Not applicable
- **IMDG**
  - •**Marine pollutant** : Not applicable
- **Air transport(IATA)**
  - •**UN No.** : Not applicable
  - •**Proper shipping name** : Not applicable
  - •**Class or division** : Not applicable
  - •**Packing group** : Not applicable

SECTION 15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

●ETC regulation - EU. Chemicals & Articles Subject to Export Ban: Annex V (Art. 15), Regulation 649/2012/EU, as amended by Regulation 2022/643, OJ L 118, 20 April 2022
- Not applicable

●ETC regulation - EU. Directive 2012/18/EU on major accident hazards involving dangerous substances, Annex I, OJ (L 197)1, 24 July 2012
- Not applicable

●ETC regulation - EU. F-Gases Subject to Emission Limits/Reporting (Annexes I, II), Regulation 517/2014/EU on FGGs, 20 May 2014

- Not applicable

●ETC regulation - EU. GHS Classification. CLP Regulation (EC) No 1272/2008, Annex VI, Table 3, Harmonized List of Hazardous Substances, as amended by Regulation (EU) 2022/692, OJ L 129, 3 May 2022

- Not applicable

●ETC regulation - EU. Polluting Substances: Annex II, Directive 2010/75/EU on Industrial Emissions (IPPC), 17 December 2010

- Not applicable

●ETC regulation - EU. REACH, Annex XIV, Substances Subject to Authorization (Authorization List), as amended through Regulation (EU) 2022/586, 11 April 2022

- Not applicable

●ETC regulation - EU. REACH, Annex XVII, Restrictions on manufacture, placing on the market and use of certain dangerous substances, 1907/2006/EC, as amended by Reg 2021/2030/EU, 22 Nov 2021

- Not applicable

●ETC regulation - EU. Regulation 1005/2009/EC on substances that deplete the ozone layer, Annex I, Controlled Substances (L286, Vol. 52, 31 October 2009)

- Not applicable

●ETC regulation - EU. Regulation 1005/2009/EC on substances that deplete the ozone layer, Annex II, New Substances (L286, Vol. 52, 31 October 2009)

- Not applicable

●ETC regulation - EU. Regulation 2019/1021/EU on persistent organic pollutants (POPs) (recast), 25 June 2019, as amended by Regulation 2021/277, 23 February 2021

- Not applicable

●Global Inventory - EU. European Inventory of Existing Commercial Chemical Substances (EINECS)

- 1,4-Benzenedicarboxylic acid bis(2-ethylhexyl) ester

- 2-Ethylhexanol

15.2 Chemical Safety Assessment

No Chemical Safety Assessment has been carried out for this substance/mixture by the supplier.

SECTION 16. Other information

16.1 Key literature references and sources for data

- DFGMAK-Doc 20 (2003), ECETOC TR48 (1998)
- Directive 84/449/EEC, C1, GLP, IUCLID
- Directive 84/449/EEC, C5, GLP, IUCLID
- Directive 87/302/EEC, C. 2, GLP, IUCLID
- EU CLP
- HSDB
- ICULID
- IUCLID
- NLM,THOMSON
- National Library of Medicine(NLM)(http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?CHEM)

- National Library of Medicine/Hazardous Substances Data Bank(NLM/HSDB)(http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?HSDB)
- PATTY (5th, 2001), JFCA 786(1993), DFGMAK-Doc.20(2003)
- TOMES;RTECS
- DFGMAK-Doc.20 (2003)
- Directive 84/449/EEC, C. 2, GLP, IUCLID

16.2 Issuing date : 2023-05-09

16.3 Indication of changes

Revision number : 0

Revision date : 2023-05-09

Revision history :

16.4 Abbreviations and acronyms